

Applied ML TA 2026

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Outline

- 1 Introduce Myself
- 2 What You May Learn from This Course
- 3 TAs Job
- 4 Common Questions
- 5 Q&A

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About Po-Yu Yeh

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Current:

- Research Assistant @ NTNU STDS
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- Chief Cybersecurity Officer @ AdaKrei Tech.
- Research Assistant @ NTU Hospital

Keywords: Quantum geography, GeoAI, Humanity GIS

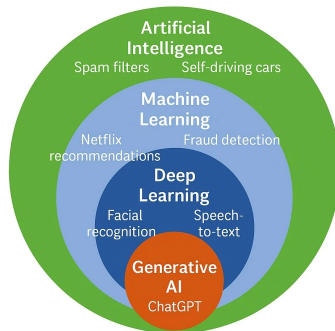
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What You Will Learn?

What is machine learning

The broadest definition of machine learning is to make a mathematical model to fit a specific distribution.



Source: freeCodeCamp

What You Will Learn?

Natural Language Processing

In this part, we want to know how to make mathematical model learn something from texts. We will start from some classic feature extraction algorithms like tf-idf to modern approach like BERT.

Large Language Model

In this part, we will go through some chatGPT-like models and its application. Unfortunately, the length of this topic is only three hours. So, we cannot

Reinforcement Learning

Reinforcement learning is a technique to make model learn how to do something. Also unfortunately, we will only go through some typical algorithms and cool applications.

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TAs Jobs

In classes...

- NLP Courses
- RL Course
- GNN Course ((Maybe...
- Your homeworks

Out of classes...

- Your questions
- fix some unsolved bugs

I will not do...

- Be a babysitter on classes
- Finetune your model
- Answer **every** problem from you guys
- ...

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What will homework look like?

The homework has basic (60-80%) and advanced (40-20%) parts. In the basic part, you will do some really **basic** jobs like preprocessing, training model, and using pretrained models. In advanced part, you may need to finetune a model, implement model from paper, or do something special upon the topic.

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Do I need my own GPU?

Not really. Theoretically, the basic part of homework can be done solely on CPU. For advanced part, I will make sure you can done it via google colab. If not, TA can help you to train your model.

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What should my final project look like?

There is no restrict about final project. Just anything about ML like applying ML into a new dataset, a new model, or a brand new ML algorithm. Everything upon you. The key to get high grade is the novelty of your project.

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How can I contact TA?

You can contact TA via email (poyu.py@ntnu.edu.tw). (Line, Instagram, Discord, or other social media are prohibited if you already have any)

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TA Time?

Because of some issues, it is pretty hard to have a physical TA time at a specific time. So, you may need to reserve your own TA time and we will meet on Google Meet or somewhere if you needed.

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End

